

# CSE105/209A: Modern Algorithmic Toolbox\*, PSET-2

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## 1

1.

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (1)$$

subtract  $\lambda$  from diagonal

$$A - \lambda I = \begin{pmatrix} -\lambda & 1 & 0 & 0 \\ 1 & -\lambda & 1 & 0 \\ 0 & 1 & -\lambda & 1 \\ 0 & 0 & 1 & -\lambda \end{pmatrix} \quad (2)$$

the characteristic polynomial is

$$\lambda(\lambda(\lambda^2 - 1) - \lambda) + (-1(\lambda^2 - 1)) = \lambda^4 - 3\lambda^2 + 1 = (\lambda^2 - \lambda - 1)(\lambda^2 + \lambda - 1) \quad (3)$$

which gives roots of

$$\begin{aligned} \lambda_1 &= -\frac{1}{2} - \frac{\sqrt{5}}{2} \\ \lambda_2 &= \frac{1}{2} - \frac{\sqrt{5}}{2} \\ \lambda_3 &= -\frac{1}{2} + \frac{\sqrt{5}}{2} \\ \lambda_4 &= \frac{1}{2} + \frac{\sqrt{5}}{2} \end{aligned} \quad (4)$$

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\*PI: Vaggos Chatziafratis

to find eigenvalues, find the null space after subtracting  $\lambda I$  from the diagonal of  $A$ .

$$(A - \lambda I)x = 0 = \begin{pmatrix} -\lambda & 1 & 0 & 0 \\ 1 & -\lambda & 1 & 0 \\ 0 & 1 & -\lambda & 1 \\ 0 & 0 & 1 & -\lambda \end{pmatrix} x = 0 \quad (5)$$

$$\begin{aligned} -\lambda x_1 + x_2 &= 0 \\ x_1 - \lambda x_2 + x_3 &= 0 \\ x_2 - \lambda x_3 + x_4 &= 0 \\ x_3 - \lambda x_4 &= 0 \end{aligned} \quad (6)$$

if we set  $x_1$  to 1,

$$\begin{aligned} x_2 &= \lambda \\ x_3 &= \lambda^2 - 1 = (\lambda - 1)(\lambda + 1) \\ x_4 &= \lambda - 1/\lambda \end{aligned} \quad (7)$$

note

$$\left[ \frac{1}{2}(\pm 1 \pm \sqrt{5}) - 1 \right] \left[ \frac{1}{2}(\pm 1 \pm \sqrt{5}) + 1 \right] = \frac{1}{2}(\mp 1 \mp \sqrt{5}) - 1 \quad (8)$$

$$\left[ \frac{1}{2}(\pm 1 \mp \sqrt{5}) - 1 \right] \left[ \frac{1}{2}(\pm 1 \mp \sqrt{5}) + 1 \right] = \frac{1}{2}(\mp 1 \pm \sqrt{5}) - 1 \quad (9)$$

$$\left[ \frac{1}{2}(1 \pm \sqrt{5}) - 1 \right] - \frac{1}{\left[ \frac{1}{2}(1 \pm \sqrt{5}) - 1 \right]} = 1 \quad (10)$$

$$\left[ \frac{1}{2}(-1 \pm \sqrt{5}) - 1 \right] - \frac{1}{\left[ \frac{1}{2}(-1 \pm \sqrt{5}) - 1 \right]} = -1 \quad (11)$$

for  $\lambda_1$ ,

$$\begin{aligned}
v_1 &= \left(1, -\frac{1}{2} - \frac{\sqrt{5}}{2}, \frac{1}{2} + \frac{\sqrt{5}}{2}, -1\right) \\
v_2 &= \left(1, \frac{1}{2} - \frac{\sqrt{5}}{2}, -\frac{1}{2} + \frac{\sqrt{5}}{2}, 1\right) \\
v_3 &= \left(1, -\frac{1}{2} + \frac{\sqrt{5}}{2}, \frac{1}{2} - \frac{\sqrt{5}}{2}, -1\right) \\
v_3 &= \left(1, \frac{1}{2} + \frac{\sqrt{5}}{2}, -\frac{1}{2} - \frac{\sqrt{5}}{2}, 1\right)
\end{aligned} \tag{12}$$

2.

$$B = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix} \quad (13)$$

$$B - \lambda I = \begin{pmatrix} 1-\lambda & -1 & 0 & 0 \\ -1 & 2-\lambda & -1 & 0 \\ 0 & -1 & 2-\lambda & -1 \\ 0 & 0 & -1 & 1-\lambda \end{pmatrix} \quad (14)$$

the characteristic polynomial of B is

$$(1-\lambda)((2-\lambda)((2-\lambda)(1-\lambda)-1)-(1-\lambda))+((-2-\lambda)(1-\lambda))+1) \quad (15)$$

which is equal to  $x(x-2)(x^2-4x+2)$  which gives

$$\begin{aligned} \lambda_0 &= 0 \\ \lambda_1 &= 2 - \sqrt{2} \\ \lambda_2 &= 2 \\ \lambda_3 &= 2 + \sqrt{2} \end{aligned} \quad (16)$$

for eigenvectors, we know  $v_0 = (1, 1, 1, 1)$

we can also guess  $v_2 = (1, -1, -1, 1)$  and it verifies to be correct.

the final two we can also guess, the diagonal is  $1 \pm \sqrt{2}, \pm\sqrt{2}, \pm\sqrt{2}, 1 \pm \sqrt{2}$

$$v_1 = (-1, 1 - \sqrt{2}, -1 + \sqrt{2}, 1) \quad (17)$$

$$v_3 = (-1, 1 + \sqrt{2}, -1 - \sqrt{2}, 1) \quad (18)$$

3.

$$C = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix} \quad (19)$$

one can guess the main eigenvector  $(1, 1, 1, 1)$  with eigenvalue 3.

we can also reason that the remaining 3 eigenvalues must be -1. a vector that does not have eigenvalue 3 must be orthogonal to the all ones vector. suppose the eigenvector is orthogonal, its components must sum to 0. the matrix  $C$  will take each component  $i$  and map it to the sum of the other components which must equal the negative of the component. Thus, any

vector that is orthogonal to  $(1, 1, 1, 1)$  must be an eigenvector of with eigenvalue -1.

Here is a valid set of eigenvectors for eigenvalues 1 through 3.

$$\begin{aligned} v_1 &= (1, 0, 0, -1) \\ v_2 &= (0, 1, -1, 0) \\ v_3 &= (1, -1, 0, 0) \end{aligned} \tag{20}$$

note, these were choosen because they span the remaining 3 dimensions and their components sum to 0. They are not necessarily orthogonal, but they can be made orthogonal since spanning vectors can be converted to orthonormal spanning vectors.

4.

$$D = \begin{pmatrix} 3 & -1 & -1 & -1 \\ -1 & 3 & -1 & -1 \\ -1 & -1 & 3 & -1 \\ -1 & -1 & -1 & 3 \end{pmatrix} \tag{21}$$

we know the all 1s vector has eigenvalue 0.

we can use a similar reasoning to the last question.

any eigenvector with eigenvalue nonzero (which we know all eigenvectors must be nonzero other than  $\lambda_0$ ) must able to be expressed as orthogonal vectors to the all ones vector, i.e. the components must sum to 0. take  $v$  to be an arbitrary vector orthogonal to the all ones.  $(Cv)_i$  for any component  $i$  is equal to  $3v_i - \sum_{j \neq i} v_j = 4v_i$ . thus any  $v$  orthogonal to the all ones is an eigenvector for  $D$ .

$$\begin{aligned} v_1 &= (1, 0, 0, -1) \\ v_2 &= (0, 1, -1, 0) \\ v_3 &= (1, -1, 0, 0) \end{aligned} \tag{22}$$

eigenvectors remain the same on complete graphs of  $n$  nodes, that being  $n - 1$  eigenvectors are eigenvalue  $n$ .

## 5. adjacency

we can solve the two block matrices seperately since the vectors this matrix acts on can be broken into two orthonormal spaces.

for the complete graph, we know

$$\lambda_0 = 0, \lambda_{2,3} = -1$$

$$\begin{aligned}
v_0 &= (1, 1, 1, 0, 0, 0) \\
v_1 &= (1, 0, -1, 0, 0, 0) \\
v_2 &= (-1, 2, -1, 0, 0, 0)
\end{aligned} \tag{23}$$

for the path graph,

$$F_a = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \tag{24}$$

we can guess

$v_3 = (0, 0, 0, 1, 0, -1)$  which has eigenvalue 0.

there is also  $v_4 = (0, 0, 0, 1, \sqrt{2}, 1)$  which has eigenvalue  $\sqrt{2}$ . (if you guess 1,2,1 you see that the output is 2,2,2 so scaling down the center element gives you the eigenvector)

the final eigenvector is  $v_5 = (0, 0, 0, 1, -\sqrt{2}, 1)$  since we know how polynomials split, the eigenvalue must include  $\lambda_5 - \sqrt{2}$  so from this we find the eigenvector.

### laplacian

we know  $v_0, v_1 = (1, 1, 1, 0, 0, 0), (0, 0, 0, 1, 1, 1)$  w eigenvalue 0 since they are disconnected.

we also know that there exists two eigenvectors w eigenvalue 3 in the  $G_1$  component

$$v_2, v_3 = (1, 0, -1, 0, 0, 0), (-0.5, 1, -0.5, 0, 0, 0)$$

the final two can be found by finding the eigenvectors for the smaller 3x3 again, we can guess for the matrix

$$F_l = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix} \tag{25}$$

$$\begin{aligned}
v_4 &= (0, 0, 0, 1, 0, -1) \\
v_5 &= (0, 0, 0, 1, -2, 1)
\end{aligned} \tag{26}$$

$$\lambda_4 = 1, \lambda_5 = 3$$

## 2

consider the graph consisting of two cliques of size 3 and one of size 2.

assume vertex 1 to 3, 4 to 6, 7,8 are connected. we can write this as a block matrix with the diagonal being 3, 3, 3, 3, 3, 2, 2 and the off diagonals in each block matrix being  $-1$ .

the eigenvectors for this that have eigenvalue 0 would be

$$\begin{aligned} v_0 &= (1, 1, 1, 0, 0, 0, 0, 0) \\ v_1 &= (0, 0, 0, 1, 1, 1, 0, 0) \\ v_2 &= (0, 0, 0, 0, 0, 0, 1, 1) \end{aligned} \tag{27}$$

we know that since the matrix is block diagonal, we chose the vectors carefully so that the product between the vector and the matrix is only affected by a specific block matrix. since each block matrix is like its own separate graph, we know the all 1s vectors on each connected component is an eigenvector with eigenvalue 0.

## 3

1. instantiate an  $n \times n$  matrix  $A$  of all zeros.
2. for each node  $i$ , choose two distinct nodes  $j, k$  that  $i$  is connected to. increment the  $A_{j,k}$  by 1.
3. if any entry in  $A$  is greater than 2, then there exists a cycle.

### runtime:

each item above takes  $n^2$  time and are not nested thus the runtime is  $n^2$ .

### correctness:

$A_{i,j}$  can only have an entry greater than two iff at least two distinct vertices  $\alpha, \beta$  are neighbors to both  $i, j$ . This means there is a 4 cycle on  $i, \alpha, j, \beta$ .

on vertex  $a$ , each loop searches for all possible wedges with center point  $a$  and increments the entry corresponding to the endpoints of the wedge. In the language of loop invariants, after each loop in item 2, the value of  $A_{i,j}$  counts the number of wedges with endpoints  $i, j$  that can be formed using only vertices before the index of the loop.

in the end, since every possible center point of wedges was checked,  $A$  exactly counts the number of wedges with endpoints  $i, j$ . two wedges with distinct center points must mean there is a cycle of length 4.

## 4

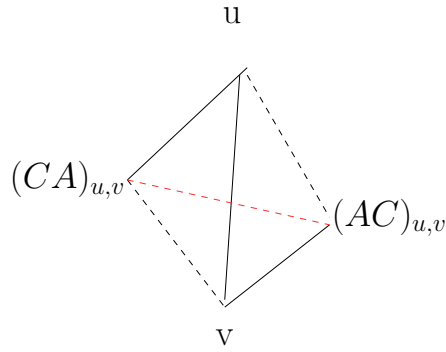
note

$$(CA)_{i,j} = \sum_k C_{i,k} A_{k,j} \quad (28)$$

this value counts how many "paths missing one edge" that end on  $j$  there are.

$$(AC)_{i,j} = \sum_k C_{i,k} A_{k,j} \quad (29)$$

this value counts how many "paths missing one edge" that start on  $j$ .



assume the red dotted line is not an edge. in this case, this motif (path on 4 vertices) gets counted exactly once with the  $uv$  edge.

if the red dotted line is an edge, each edge contributes once to the sum, which is why cycles on 4 nodes get counted 4 times.

i.e. the cycle has 4 paths of length 3 NOT up to isomorphism and since the equation counts paths of length 3, it each cycle shows up 4 times.

## 5